FINAL PROJECT REPORT

Multimodal Product Classification for Large-Scale E-Commerce

Data audit, text models, image fine-tuning and late fusion on the Rakuten France catalogue

Key results

- **Final model:** a Random Forest on the probabilities of CamemBERT, the SVM and ConvNeXt reaches **0.8988 weighted F1** (**0.8856 macro F1**). That is **+2.22 points** over the best text model and **+19.89 points** over the best image model.
- **Per category:** the fusion is better than CamemBERT alone on **26 of the 27 categories**.
- **Text:** fine-tuned **CamemBERT** reaches **0.8765 weighted F1**, with 22% fewer errors than TF-IDF + LinearSVC (0.8430).
- **Image: ConvNeXt-Tiny** with smart white-border cropping and layer-wise learning rates reaches **0.6998 weighted F1**, against 0.4817 for the frozen ResNet18 baseline.

Dataset: Rakuten France Multimodal Challenge
Scope: 84916 products, 27 categories
Authors: Data Science & Machine Learning Team

Evaluation: fixed stratified 80/20 split (16984 validation products)
Models: TF-IDF + SVM, CamemBERT, ConvNeXt-Tiny, Random Forest fusion
Date: November 2026

1. Summary and Problem

Marketplaces receive millions of product listings from sellers. Each listing must be placed in the right category, or search and browsing suffer. The **Rakuten France Product Classification Challenge** asks to classify listings into **27 categories** (`prdtypecode`) from their text (title and description) and their image.

84916	27	35.1%	0.8988
PRODUCTS	CATEGORIES	MISSING DESCRIPTIONS	FINAL WEIGHTED F1

Text is the strongest single source. It fails when titles are short, generic or misleading. Images help in exactly those cases, mainly for books, where the cover shows whether a book is new, used or part of a lot. We combine a TF-IDF + SVM model, a fine-tuned CamemBERT and a fine-tuned ConvNeXt image model. A Random Forest learns how to combine their probabilities.

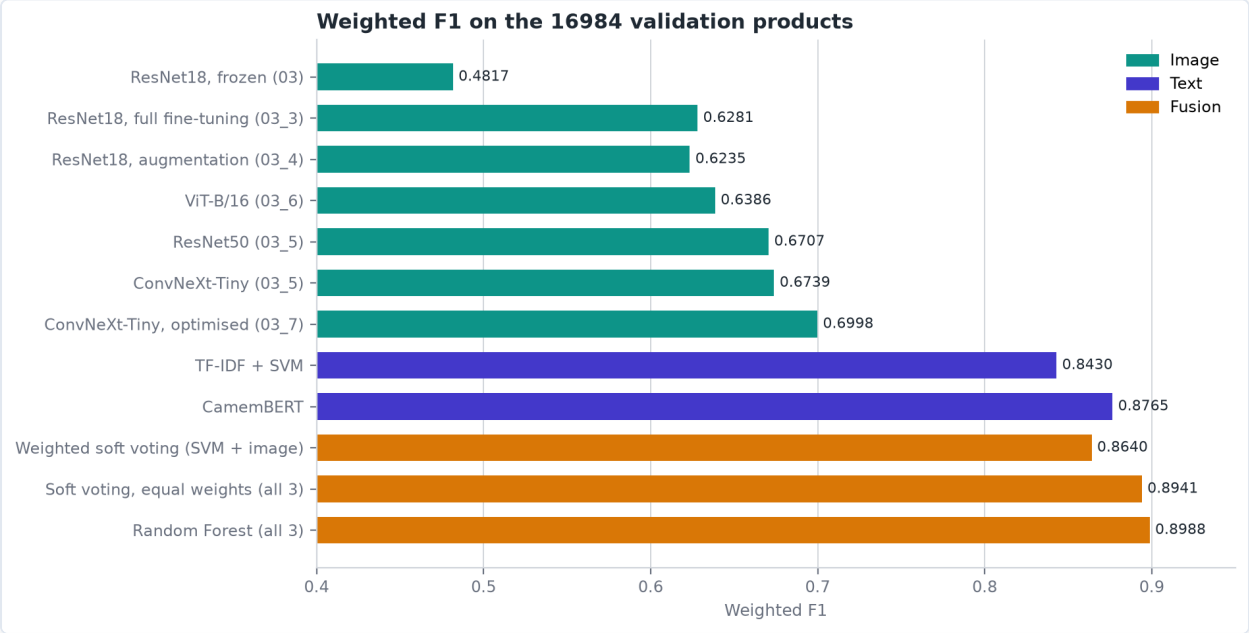


Figure 1: *Weighted F1 of the image models, the text models and the fusion methods, on the same 16984 validation products.*

2. Data Audit

2.1 Class Imbalance

The labelled data contains **84916 products** in 27 categories. The categories are very unbalanced:

- The largest category, **Pool (code 2583)**, has **10209 products** (12.0%).
- The smallest category, **Wargames (code 1180)**, has **764 products** (0.9%).
- The ratio is **13.4 to 1**. Accuracy alone would hide errors on small categories, so we report **weighted F1** (the challenge metric) and **macro F1** (every category counts the same).

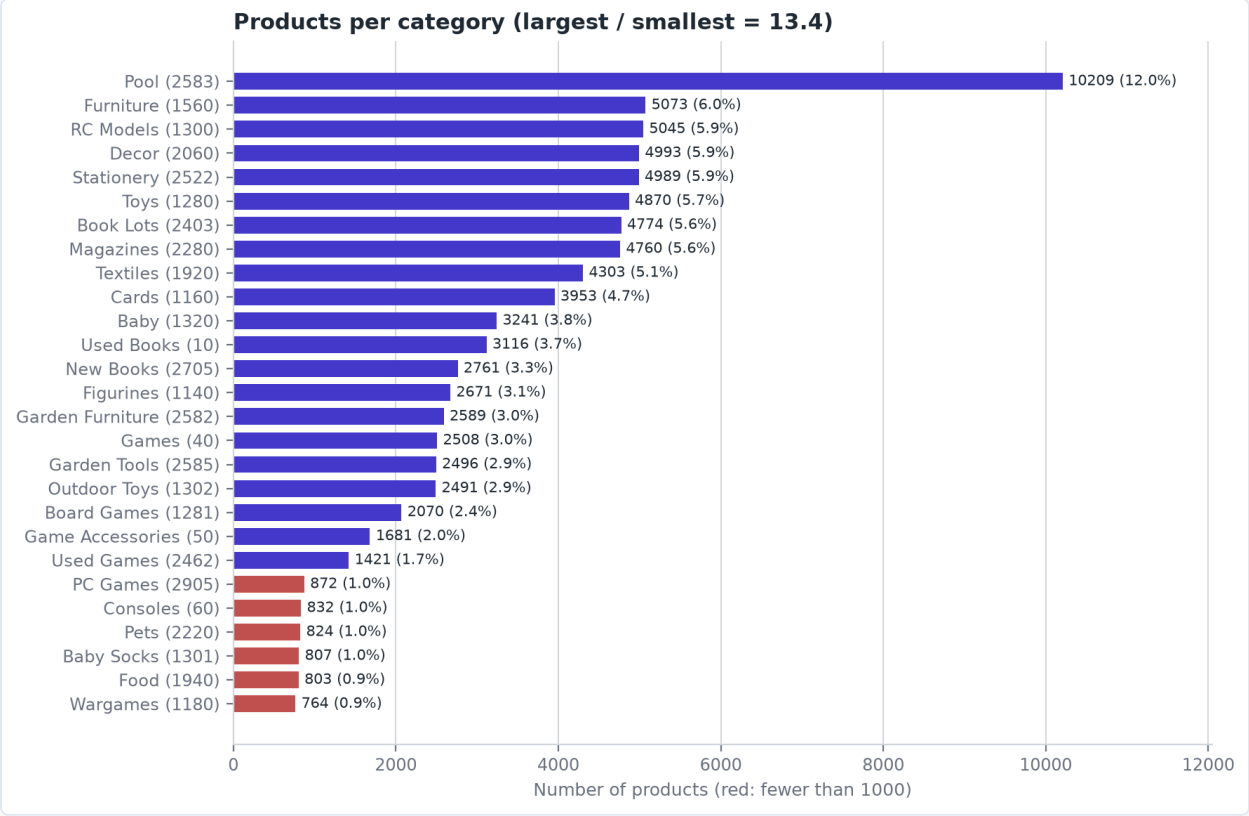


Figure 2: *Number of products per category. Seven categories have fewer than 1000 products.*

2.2 Text: Title and Description

- **Titles:** every product has a title (`designation`). The median title has 11 words.
- **Descriptions: 29800 products (35.1%)** have no description. In Book Lots (2403), 97.4% are missing. For these products, the model relies on the title and the image.
- **HTML noise:** encoded HTML characters (such as `&`, `é`) appear in 41.3% of descriptions, and HTML tags in 18.4%.
- **Repeated titles:** 4187 products (4.9%) share a title with another product. 160 of these groups contain more than one category.

2.3 Images

- **Coverage:** every product has an image. The sampled images are all 500 × 500 pixel RGB JPEGs.
- **White borders:** many images show a small product in the middle of a large white background. Resizing the full image makes the product very small.
- **Duplicates:** some different products share the exact same image file (for example placeholder images). They are found with a SHA-256 hash of the file.

3. Preprocessing and Shared Split

> **One split for all models:** a single stratified **80/20 train/validation split** (67932 training products, 16984 validation products) is used by every text, image and fusion model. Settings are chosen on the training part only. The validation part is used for the final scores.

3.1 Text

1. **HTML decoding:** entities such as `é` and `&` are turned back into characters (*é*, *&*).
2. **Tag removal:** HTML tags are removed, with a space in their place so words do not merge.
3. **Lowercase and spaces:** text is lowercased and repeated spaces are collapsed. Numbers, units and model codes are kept.
4. **Combination:** title and description are joined into one text (`text_combined`). If there is no description, only the title is used.

3.2 Images

Images stay in their original files. A manifest (`image_split_clean.csv`) lists each image with its split and the SHA-256 hash of its file. Training images whose file also appears in validation are removed, so the same image cannot be in both parts. The white-border crop is computed when the image is loaded.

4. Text Models

4.1 Choosing the Classifier

The text is turned into TF-IDF vectors of words and word pairs (binary weighting, all features). Three classifiers were compared with 3-fold cross-validation on the training part:

Classifier	CV weighted F1	Observation
Complement Naive Bayes	0.7600	Fast, but 7 points below the SVM
Logistic Regression (SGD)	0.8084	Gives probabilities, 2 points below the SVM
LinearSVC	**0.8284**	**Best; kept as the classic text model**

4.2 Tuning the SVM

- **No feature limit:** there are 460326 words and word pairs. The baseline kept only 50000. Using all of them adds 1.2 points (cross-validation on train).
- **Binary TF-IDF:** a word counts once, however often it is repeated. This stops sellers' repeated keywords from dominating. It adds 0.5 points, more than the cross-validation spread (0.3).
- **Regularisation:** a grid search over C (0.1, 0.5, 1, 2) chose C = 1. Only C = 0.1 is clearly worse.
- **Probabilities:** LinearSVC gives scores, not probabilities. `CalibratedClassifierCV` (3 folds, sigmoid) turns them into probabilities for the fusion.

On validation, the tuned SVM reaches **0.8436 weighted F1**, against 0.8313 for the baseline. With calibration it reaches **0.8430** (macro F1 0.8326).

4.3 CamemBERT

To read words in context, we fine-tuned **CamemBERT** (`almanach/camembert-base`, 110 million parameters), a French language model trained on the French part of the OSCAR web corpus.

> **Training setup:** 2 epochs on the 67932 training texts, batch size 16, AdamW with learning rate 5×10^{-5} and weight decay 0.01, linear warmup over the first 6% of steps, then linear decay. Texts are cut at 128 tokens.

- **Epoch 1:** training loss falls from 2.64 to 0.58. Validation loss 0.501, weighted F1 **0.8527**.
- **Epoch 2:** training loss ends at 0.34. Validation loss 0.427, weighted F1 **0.8765**, macro F1 **0.8594**.
- Validation loss still falls in epoch 2, so the model is not overfitting.

CamemBERT against the SVM

CamemBERT makes **2082 errors** against **2656** for the SVM (21.6% fewer). The largest gains are on the book categories: Used Books +12.3 points, Magazines +10.9, New Books +10.7, Book Lots +10.0. The SVM stays better on Wargames (–8.4 points for CamemBERT), where rare exact words such as *warhammer* and *heroclix* decide.

5. Image Models

Classifying products from images alone is hard: packaging looks alike and many images have large white borders. Over several experiments, the image model moved from a frozen ResNet18 to a fully fine-tuned ConvNeXt-Tiny.

Experiment	Backbone	Main change	Weighted F1	Macro F1
Baseline (03)	ResNet18	Frozen backbone, only the last layer trained	0.4817	0.4230
Full fine-tuning (03_3)	ResNet18	All layers trained	0.6281	0.5943
Regularisation (03_4)	ResNet18	Augmentation and regularisation	0.6235	0.5876
Vision Transformer (03_6)	ViT-B/16	Self-attention on 16 × 16 image patches	0.6386	0.5982
Stronger CNNs (03_5)	ConvNeXt-Tiny	Modern CNN backbone	0.6739	0.6458
Optimised ConvNeXt (03_7)	**ConvNeXt-Tiny**	**Smart white-border crop, layer-wise learning rates, balanced sampling, flip TTA**	**0.6998**	**0.6752**

5.1 What Makes the Final Image Model Better

1. **Smart white-border crop:** if the product covers less than 75% of the image, a square window around the product is cut out at its original resolution (at least 224 pixels). The product is no longer shrunk with its white border. Photos that fill the frame are left unchanged.
2. **Layer-wise learning rates:** the whole network is trained, with four learning rates: early layers 1×10^{-5}, stage 3 3×10^{-5}, stage 4 8×10^{-5}, head 1×10^{-4}, with cosine annealing.
3. **Balanced sampling and TTA:** a weighted sampler (power 0.25) shows small categories more often. At prediction time, the image and its mirror image are averaged (test-time augmentation). This raises weighted F1 from 0.6980 to 0.6998.

6. Text and Image Fusion

Text and images have different strengths: text names the product and its details, while the image shows its shape and condition. We combine the models at the level of their predicted probabilities (late fusion).

6.1 Fusion Methods

For each of the 16984 validation products, every model gives 27 probabilities, one per category. None of the models was trained on these products.

- P_{SVM}: calibrated probabilities of the tuned LinearSVC (weighted F1 0.8430).
- $P_{\text{CamemBERT}}$: softmax probabilities of CamemBERT (0.8765).
- P_{ConvNeXt}: softmax probabilities of the optimised ConvNeXt-Tiny (0.6998).

Every fusion method is scored with 5-fold cross-validation on the validation products: it is set up on 4/5 of them and predicts the remaining 1/5.

1. **Soft voting:** average the probabilities, then take the category with the highest average.
   - *Equal weights:* each of the three models counts 1/3.
   - *Weighted:* each model gets its own weight (steps of 0.1, summing to 1), chosen on the 4/5 part.
2. **Random Forest (stacking):** the three probability vectors are joined into 81 inputs ($[P_{\text{CamemBERT}} \,\|\, P_{\text{SVM}} \,\|\, P_{\text{ConvNeXt}}]$). A Random Forest (300 trees) learns from them which model to trust for which category.

6.2 Results

Model	Input	Weighted F1	Macro F1	Gain vs CamemBERT	Errors
ConvNeXt-Tiny	Image	0.6998	0.6752	−17.67	5043
Calibrated LinearSVC	Text	0.8430	0.8326	−3.35	2656
Weighted soft voting (SVM + image)	Fusion	0.8640	0.8548	−1.25	2278
CamemBERT	Text	0.8765	0.8594	Reference	2082
Weighted soft voting (CamemBERT + image)	Fusion	0.8882	0.8730	+1.16	1875
Weighted soft voting (all 3)	Fusion	0.8920	0.8797	+1.54	1813
Soft voting, equal weights (all 3)	Fusion	0.8941	0.8826	+1.76	1776
Random Forest (all 3)	Fusion	**0.8988**	**0.8856**	**+2.22**	**1715**

- **Soft voting already gives most of the gain:** equal weights on the three models reach 0.8941.
- **Choosing the weights does not help:** weighted soft voting (0.8920) is slightly below equal weights. The weights chosen on 4/5 of the products do not transfer exactly to the last 1/5.

- **The Random Forest adds 0.5 points more** (0.8988), because it can trust a different model for each category.

6.3 Per Category

- **Against CamemBERT:** the Random Forest is better on **26 of the 27 categories**. Only Baby Socks (1301) drops slightly (–0.3 points).
- **Books:** the largest gains are New Books (+10.8 points) and Used Books (+9.9). The cover image shows whether a book is new or used, which the text often does not say.
- **Wargames (1180):** CamemBERT's weakest category (F1 0.614) rises to 0.706 (+9.3 points). The SVM catches the rare exact words (*warhammer*, *heroclix*).
- **Against weighted soft voting:** the Random Forest makes 98 fewer errors and is better on 18 of 27 categories, mostly books (New Books +4.2, Used Books +3.1).

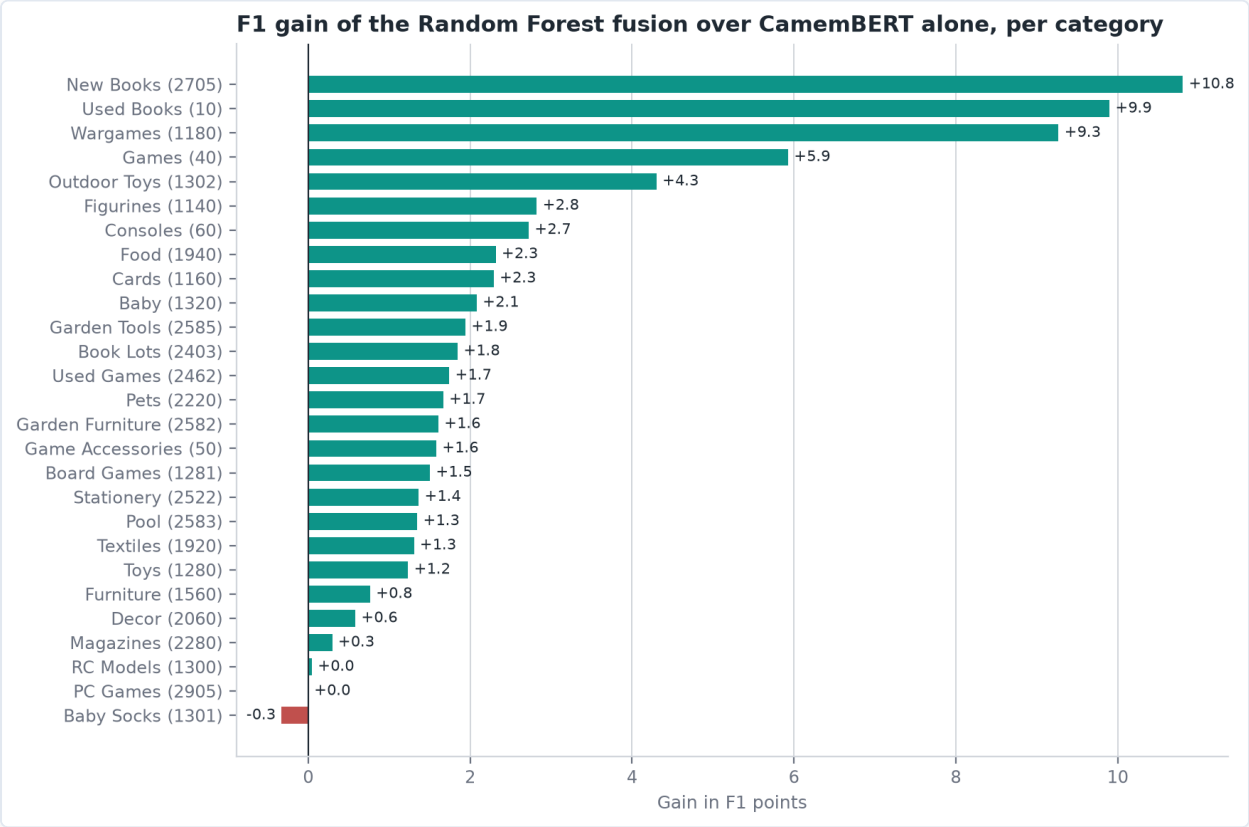


Figure 3: *F1 gain of the Random Forest fusion over CamemBERT alone, per category. Only Baby Socks is slightly lower.*

7. Interpretability

7.1 Text: What the SVM Relies On

- **Clear product words:** most categories have sensible top words, such as *piscine* and *intex* for Pool, *chien* and *laisse* for Pets, *warhammer* and *heroclix* for Wargames.
- **Seller names and writing style:** some top words are seller or brand names (*sweetheart93*, *cdaffaires*). New Books relies on words from book summaries (*ce livre*, *auteur*). This works on this data but may not hold for new sellers.
- **Why books get confused:** each book category has a few strong words (*lot* and *tomes* for Book Lots, dates for Magazines). Common book words such as *roman* have almost the same weight in all four. A short book title without a strong word gives the model little to decide with.

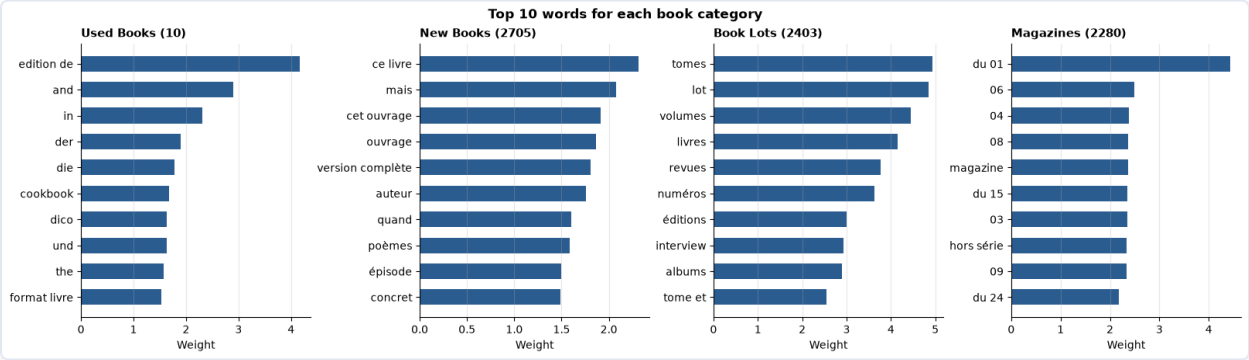


Figure 4: *Highest SVM weights for the four book categories. Each category has a few strong words; common book words are shared.*

7.2 Fusion: What the Random Forest Relies On

- **Importance:** the Random Forest relies mostly on CamemBERT (about two thirds of the importance), then on the SVM (23%), then on the image (11%).
- **Image contribution:** without the image, the fusion scores 0.8873. With it, 0.8988 (+1.2 points).
- **Where the image matters:** it adds almost all of its gain on New Books (+11.4 points) and Used Books (+10.1), and helps on Games (+4.3). On most other categories it adds little.

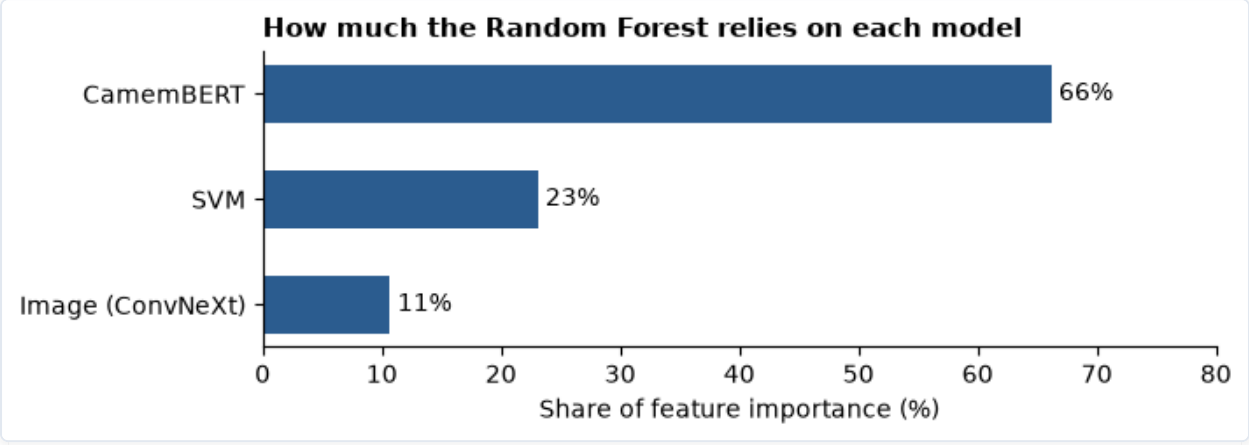


Figure 5: *Share of the Random Forest importance that comes from each model's probabilities.*

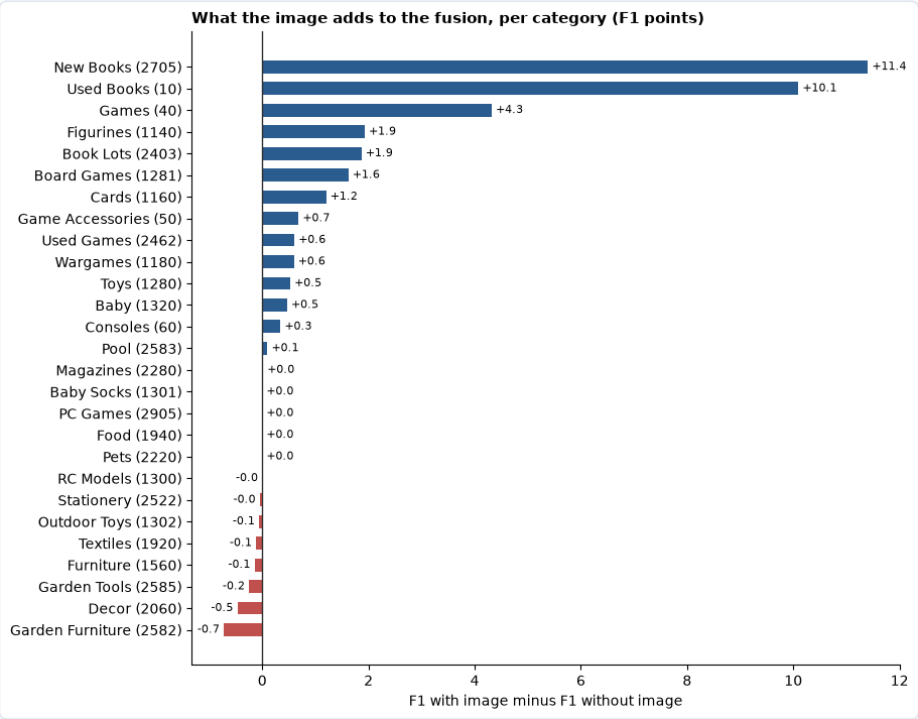


Figure 6: *F1 of the Random Forest with the image minus without the image, per category (same 5-fold cross-validation).*

8. Recommendations and Conclusion

8.1 Which Model to Use

Need	Model	Weighted F1	What it needs
Cheap and fast	Calibrated LinearSVC (text only)	0.8430	CPU only; trains in minutes.
Simple fusion	Soft voting, equal weights (all 3)	0.8941	The three models; no extra model to train.
Best accuracy	**Random Forest (all 3)**	**0.8988**	The three models plus a Random Forest trained on their probabilities.

8.2 Conclusion

Text carries most of the information, and CamemBERT is the best single model. The image adds a check where text is weak, mainly for books. Combining the three models through a Random Forest gives **0.8988 weighted F1** on the 16984 validation products. Simple soft voting with equal weights comes close (0.8941) and is easier to run.

> **Reproducibility:** the notebooks (`01` to `05`), the processed data, the saved probabilities and the script that builds this report (`scripts/make_final_report.py`) are in the repository. Model checkpoints are too large for git and are shared separately.